

Foodereum: A Blockchain-based Authenticated Solution for Food Supply Chain

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Abstract—Blockchain technology has evolved and been widely available over the past several years, revolutionising a variety of industries, most notably the supply chain industry. The food supply chain currently in effect across different nations faces numerous challenges, including a lack of transparency and traceability of food items due to a centralised system, third-party interference during shipment, high transportation and storage costs, food waste due to rotting and contamination and poor accountability of entities involved. The decentralised nature of blockchain, as well as its ability to track ownership records and resist tampering, can be employed to tackle these critical challenges. This paper introduces Foodereum, a blockchain-based authenticated solution for food programmes that enables organisations to record, track and authenticate food products along the supply chain. The research also examines the industry's present issues followed by Foodereum's characteristics, a detailed analysis of gas consumption and future scope.

Index Terms—Blockchain, Ethereum, Supply chain, Traceability, Smart contract, Authentication, Food safety, Gas value, Solidity

I. INTRODUCTION

CURRENTLY, conventional records in food supply chains merely monitor and store orders and shipments, without incorporating essential features such as transparency, traceability, and auditability. Furthermore, communication gaps between supply chain partners, the possibility of food fraud, rising supply chain costs and growing regulations highlight the system's inefficiencies. Tackling these hurdles will improve food quality and safety, gaining popularity among consumers [1]. Blockchain improves supply chain efficiency by allowing organizations to conduct transactions without the intervention of third parties.

Blockchain [2], [3] is a decentralized, immutable and distributed ledger that may be used to replace the traditional method of recording transactions by employing a centralized

database with transactions that are validated by network nodes and recorded on a new block in the blockchain [4], [5], [6]. To ensure trust and reliability across the supply chain, the stored assets must be tamper-proof. Since all of the entries maintained in a blockchain are based on a consensus reached by the overall majority of the network's peers, this distributed ledger is immutable by design and provides an auditable and transparent source of information. Blockchain technology is applied in a wide range of industries, such as banking, healthcare, IoT (Internet of Things), insurance and supply chain management [7]. It has been demonstrated to be extremely successful in combating fraud and theft. As a result, blockchain reveals all of the key traits for decentralizing food authentication systems by producing traceable data that is accessible at each and every step in the supply chain.

In this paper, we present Foodereum, an Ethereum-based authentication system for food programmes that allow organizations to record, track and authenticate food products along the supply chain. Since immutability is a virtue of the blockchain, transactions cannot be tampered or hidden because every transaction is recorded in a distributed ledger, which also can be traced and revealed to the whole network who have access to the data. This implies that anytime a product is delivered, a new receiver will be able to view the product's entire history on the blockchain network. A comprehensive performance analysis of the proposed method in terms of gas consumption has been reviewed.

The remainder of this paper is organized as follows: Sec. II summarizes the related work of our problem statement; Sec. III describes the system model of our project; Sec. IV contains the implementation details like the Foodereum architecture and the General skeletal structure of blocks; Sec. V details the simulations and results of our paper and Sec. VI concludes our paper.

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II. RELATED WORK

Blockchain is a digital record of transactions that is replicated and distributed over the whole network of computer systems, which drives Bitcoin and other cryptocurrency networks and has the potential to transform the financial landscape. However, blockchain has the potential to significantly enhance supply chains by allowing faster and more cost-effective product delivery, increasing product traceability, boosting partner coordination, and facilitating access to funding.

[8] presents AgriBlockIoT, a fully decentralized traceability solution for the Agriculture and Food (Agri-Food) supply chain management. By directly creating and consuming critical information from IoT devices along the supply chain and storing such data directly in its underlying blockchain. All players (including IoT devices) must be registered users of the blockchain, having the required public/private key pairs to digitally sign each action on the distributed ledger. The proposed solution ensures transparent, immutable and auditable asset tracking in a trustless environment. However, no discussion has been made of the performance study of more limited hardware designs to test the suitability of the proposed framework to applications employing real IoT devices and gateways along the Agri-Food supply chain.

In [9], A blockchain-based reputation system is implemented in the Agri-Food supply chain. It employs Ethereum smart contracts to provide trust across trade entities and to ensure that the buyer is aware of the seller's reputation before collecting data from the seller. When entities sign a smart contract for a transaction, a reputation smart contract is initiated, which provides reviews of available data sellers. The system maintains the credibility of trade entities and product quality ratings. Furthermore, because it is built on blockchain, it ensures the immutability and integrity of the recorded review. The paper also provides extensive simulation results proving that the system requires a set amount of waiting time in order to mine the transaction. Nonetheless, the waiting time is independent of input length of reviews. However there are no discussions made about how to improve the efficiency of the proposed solution, using the performance parameters.

[10] offers a complete solution for the Agri-Food supply chain using blockchain technology. Details regarding the reputation, trade, delivery, traceability, and performance of smart contracts have also been presented. To maintain the legitimacy of the supply chain entities, a reputation system is also included. However, there is no mechanism in place to identify fake consumer reviews.

In paper [11], the proposed remedy makes use of Ethereum's blockchain and smart contracts. A general framework for tracking and carrying out commercial transactions without the use of middlemen and a centralised source of processing for product traceability throughout the agricultural supply chain is described. However, it does not address the principal difficulties in the blockchain, in terms of scalability, governance, identity registration, privacy, and legislation.

In [12], a blockchain-based system is presented for tracing

and tracking fish products along the fishing supply chain in a transparent, responsible, secure, private, and trustworthy manner. By providing transparent interactions amongst all parties, our suggested method eliminates many sorts of seafood fraud and malpractice. It has not, however, been deployed or tested on a live Ethereum network.

In [13], a method for monitoring crop prices in the agricultural supply chain is demonstrated using blockchain to regulate, track and execute business activities by eliminating intermediaries. The proposed solution includes automated payments and delivery proof, in which parties are compensated using cryptocurrency via smart contracts for successful product distribution. However, the approach does not address concerns such as identity authentication, enforcement, scalability, privacy standards and legislation. For supply chain management, the most advanced private blockchain must function in real time with product identification and location monitoring using QR codes.

III. SYSTEM MODEL

We provide our system model in this section, which is used to identify and analyse the system's key components, as well as their linkages and interactions. Our proposed model is a three-layered architecture with supply chain entities in the first layer, a smart contract and notification system in the second layer, and a simulated Ethereum blockchain in the third layer.

The shipping details which are generated from the first layer through the supply chain entities are stored in a ledger on the second layer. Through smart contract, the shipping events are then recorded on the blockchain in the third layer. The transactions which are stored in the ledger at the second layer contain the hashes of the blocks, which helps in efficient storage management.

In the proposed system, a strict control access mechanism is integrated on the basis of the roles played by the supply chain entities, in order to determine who has access to read or write the data on the blockchain. The second layer also includes a notification system, which is responsible for establishing communication and facilitating trust between entities. This feature also acts as a reputation system, which is in charge of handling discrepancies or miscommunication between entities. The third layer in the system responsible for storing the data in the blockchain which is produced from layer two.

Layer 1 consists of the several supply chain entities who communicate via the smart contract and act according to their roles. They are:

- **Sanctioner:** The only person who has authority to sanction food commodities to the Consumers. Thus, a genesis block will be mined at the time of registering the food commodities in the system.
- **Region-wise Distributors:** The distributors are in charge of authenticating the shipments at each level which tends to record accurate data upon receiving the physical package and transporting the package from place to place.
- **Consumers:** The end users who receives the food commodities.

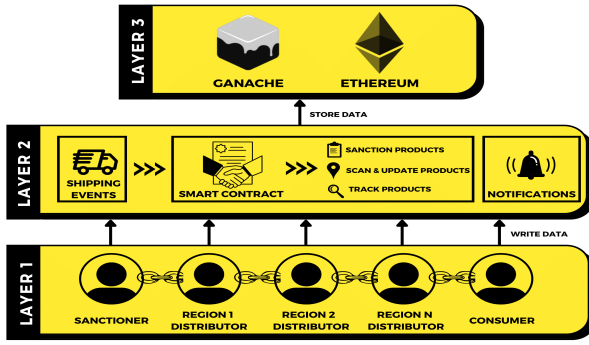


Fig. 1. System Model

Layer 2 consists of shipping events, a smart contract and a notification system. As the sanctioner registers the food commodities to the consumers, the commodities have to be shipped from place to place. In that context, the shipping events in the layer 2 are the details of every shipment which are authenticated by the distributors at their region across the supply chain. These shipping events are then recorded on the blockchain which can be accessed by the entities to track shipment status of the food commodities. A self-executing Smart contract which is present in the layer 2 of our system, contains three prominent functions which enables the entities to interact in the supply chain. The three functions are:

Sanction products (): In Sanction Product, the sanctioner can sanction food commodities as a single block (Genesis Block). The function accepts the name of the product and its quantity to be sanctioned as the input. Following sanctioning, a QR code and its associated Product ID will be generated, which may be downloaded and used to trace the product throughout the network. Once the food goods have been approved, users must be informed of the details through an inbuilt notification system. This makes tracking the food products transparent and efficient.

Scan & Update Products (): The Scan & Update Product feature allows the user to authenticate food products by updating the shipment status and the product's Geo-location by either scanning the QR code or entering the product ID. This function takes Product ID and the Geo-location as the inputs. Using QR code, the user can automatically update the shipment details without having to enter it manually. The Scan Update feature provides a cost-efficient, paper-free, and a time-saving alternative to traditional inventory management systems.

Track Products (): The user can keep tabs on the status of food products that have been updated utilising the Scan & Update Product feature by using the Track Product feature. The Product ID may be entered into this function. By scanning the QR code or inputting the Product ID, each food product may be traced in detail at every level. Details include the parties involved in prior shipments, the date, time, and place. In the event of a quality issue, the product may be quickly tracked down from its place of origin. This function guarantees

total system transparency and traceability.

A **Notification system** is also a part of second layer, which is accounted for establishing communication and enabling trust between entities. The Notification system also acts as a reputation system which is in charge of handling discrepancies or miscommunications between entities as any entity can communicate with any other entities in the chain to solve or raise issues. This feature also helps Food Sanctioners to assure the authenticity and reliability of their supplies by tracking tampering attempts as the food commodities flows through the supply chain.

At **Layer 3**, the data or the shipping events which are produced by the supply chain entities from layer two are stored in the blockchain against their roles / profiles. The transactional data which are stored in the ledger at the second layer contain the hashes which points to the blocks of the Ethereum blockchain, these transactional data are stored as metadata to overcome the limitations of a storage unit, this helps in efficient storage management. Due to the incorporation of a strict control access mechanism in the system, the access to read or write the data on the blockchain on the basis of the roles of the entities, ensures that the network's privacy and confidentiality are protected.

IV. IMPLEMENTATION DETAILS

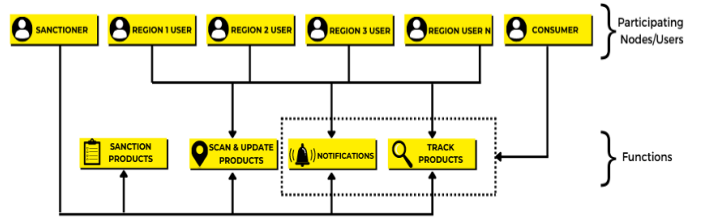


Fig. 2. Block diagram

Fig. 2 depicts how various users in the supply chain have access to different functionalities. Out of all the users in the supply chain, only the sanctioner has the authority to sanction a product. The features that the sanctioner may access are Sanction Products, Scan and Update Products, Track Products, and Notification. The distributors in Region 1 through Region N are responsible for scanning delivered products and updating data on the blockchain. This helps to authenticate each product on the blockchain and holds all users accountable for the safety of the product. The traceability and transparency in the system can be achieved by the Track Product feature, which helps any entity in the chain to view the product's complete shipment history. By using the notification feature, any entity in the chain can send a message to other entities informing them that the product is of low quality or missing, thus the notification system acts as a reputation system. Only two functions are available to the consumers: Track Product and Notification.

Fig. 3 represents the flow of data between distinct regional blocks in the blockchain. In the system, the materials flow along the supply chain in a forward direction whereas, back

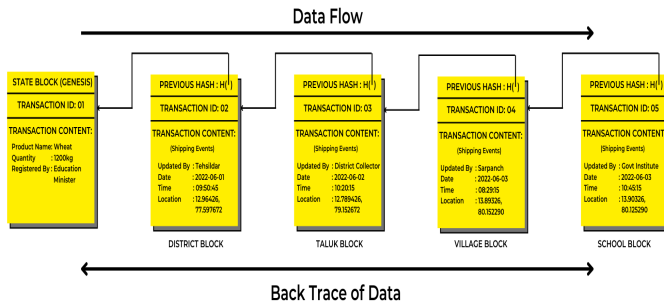


Fig. 3. Data flow diagram

trace of data is bidirectional. Each block provides the data of a specific region that the user has recorded, i.e. At each level the package is authenticated and the details are stored in blocks, which can be tracked by any entity at any level. The blocks contain information on who updated it, as well as the date, time, and location. Each block contains a distinct transaction ID that makes it easier to identify the original source of data.

V. SIMULATIONS AND RESULTS

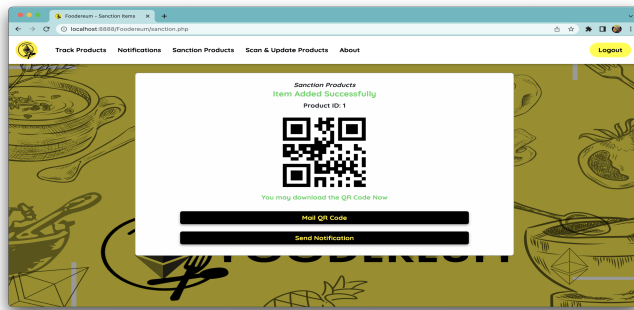


Fig. 4. Sanction Products

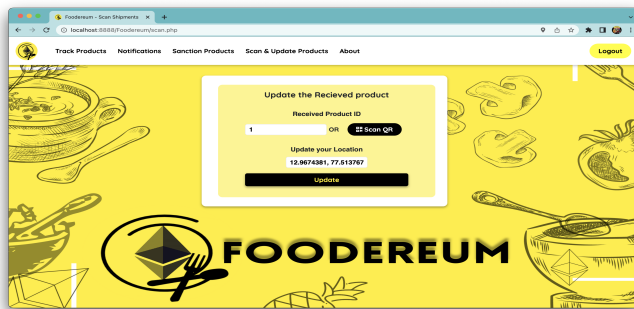


Fig. 5. Scan & Update Products

Software Requirements:

- **Ethereum:** For the simulations, an open-source blockchain technology, Ethereum, is employed. Ethereum develops decentralized apps using blockchain technology. It enables users to create smart contracts

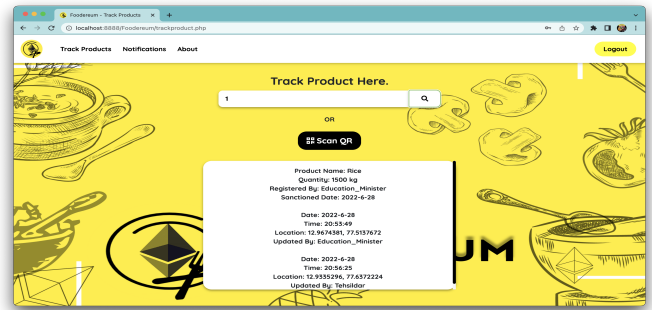


Fig. 6. Track Products

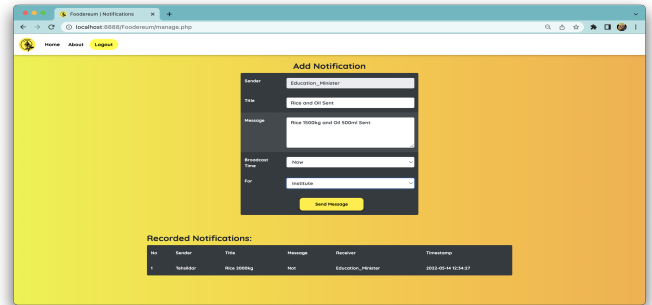


Fig. 7. Notifications Page

to form agreements without the intervention of a third party.

- **Remix Integrated Development Environment (IDE):** Remix is used to create, execute and test smart contracts for the blockchain-based supply-chain network. Solidity is the language used in remix to create smart contracts.
- **Ganache:** Ganache gives virtual accounts with a pre-defined quantity of crypto-currency in Ethers which gets deducted after each transaction. Each Ganache account has its own private key and address.
- **Metamask:** It is a browser plugin that works as a gateway between Ganache and Remix IDE, allowing them to communicate [14].
- **The system specifications are as follows:** Intel Core i5 8th Generation, 2.4 GHz processor, 8 GB RAM, and 1.5TB storage.

Analysis of gas consumption by Smart contract and its functions:

Fig. 8 depicts a detailed analysis of gas consumption in terms of Wei of the smart contract and its functions. It is clearly evident that deploying the smart contract in the Ethereum Blockchain takes the maximum value in terms of Wei, compared to any other smart contract functions [15]. Our project Foodereum consists of three different smart contract functions, which are: Sanction Products (), Scan & Update Products () and Track Products ().

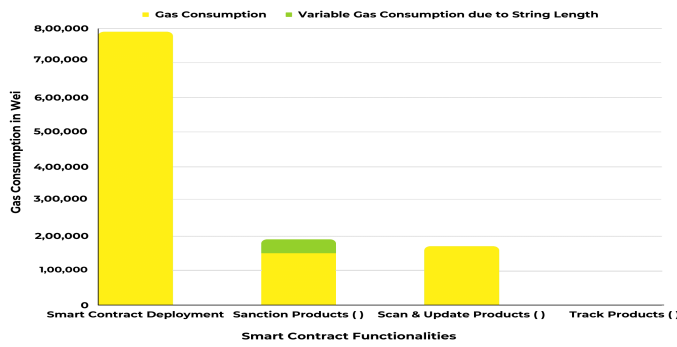


Fig. 8. Gas consumption vs Smart contract and its functions

- The Sanction products () accepts the Product name and its quantity as inputs in strings, and spends highest gas value compared to the other two smart contract functions, as this function is accounted for registering the food commodities which are sanctioned by the sanctioner in the chain, and then the products are recorded against the sanctioner's profile in blockchain. It is observed that, as the input string length increases, the amount of gas spent by the function also increases, thus there is a slight variation in gas consumption with respect to the string length.
- The Scan & Update products () spends relatively less gas compared to Sanction Products (), as this function is used while shipping the products at different regions, for which it contains the details of the products which are sanctioned earlier. There is no considerable variability involved in gas consumption when the function is triggered.
- The Track product () is the only functionality that is offered to all the entities in the supply chain, this feature tends to spend no gas in the system. As tracking plays a crucial role in a supply chain environment, and its usage can be very frequent, due to which, to view the details of sanctioned products which are record on blockchain made reliable. This feature ensures complete traceability and transparency in the supply chain.

VI. CONCLUSION AND FUTURE SCOPE

The conventional Food Supply Chain spanning different nations provides customers with access to fresh and nutritious food. The food industry plays a vital role in supplying the essentials and needs for numerous human activities. However, the present supply chain merely monitor and store orders and shipments without incorporating crucial features such as transparency, traceability, and auditability. A lack of transparency and traceability as a result of centralised systems, third-party involvement during shipment, expensive transportation and storage costs, food waste due to rotting and contamination, insufficient accountability of organisations involved, and communication gaps between supply chain partners highlight the system's inefficiencies. By overcoming these hurdles, food quality and safety will increase, gaining appeal among consumers.

By offering authentication in a real-time setting, our proposed work assists in the adoption of private blockchain for supply chain management. In order to help organisations record, track, and verify food goods along the supply chain, we propose an Ethereum-based authentication system for food programmes. Since the blockchain's immutability is one of its benefits, transactions cannot be altered or concealed since they are all recorded in a distributed ledger that can be tracked and made public to everyone with access to the data. This suggests that whenever a product is delivered, a new receiver will have access to the product's whole history on the blockchain network.

For future enhancement, we would like to enhance food product traceability and accountability through the use of Internet of Things-based sensors, cameras, RFIDs, etc. Supporting distribution across several areas would increase the scalability of the current system. Enhancing security by connecting different users with metamask accounts.

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